

Q1. (1.1) $\lim_{x \rightarrow 2} \frac{\sin(x-2)}{(x^2-4)}$

$$= \lim_{x \rightarrow 2} \frac{\sin(x-2)}{(x-2)} \times \frac{1}{x+2}$$

$$= \lim_{x \rightarrow 2} \frac{\sin(x-2)}{x-2} \times \lim_{x \rightarrow 2} \frac{1}{x+2}$$

$$= 1 \times \frac{1}{4}$$

$$= \frac{1}{4}$$

(1.2) $\lim_{x \rightarrow -\infty} \frac{\sqrt{6x^6+5}}{3x^3-10}$

$$x^3 = -\sqrt{x^6} \text{ if } x < 0$$

$$= \lim_{x \rightarrow -\infty} \frac{\sqrt{6x^6+5}}{3x^3-10} \frac{(1/x^3)}{(1/x^3)}$$

$$= \lim_{x \rightarrow -\infty} \frac{-\sqrt{(6x^6+5)/x^6}}{3 - 10/x^3}$$

$$= \lim_{x \rightarrow -\infty} \frac{-\sqrt{6 + 5/x^6}}{3 - 10/x^3}$$

$$= \frac{-\sqrt{6}}{3}$$

$$(1.3) \quad f(x) = \begin{cases} \frac{x-1}{x^2-1} & \text{if } x > 1 \\ \frac{x}{x+1} & \text{if } x < 1 \end{cases} \quad (2)$$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} \frac{x}{x+1} = \frac{1}{2}$$

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} \frac{x-1}{x^2-1} = \lim_{x \rightarrow 1^+} \frac{1}{x+1} = \frac{1}{2}$$

$$\Rightarrow \lim_{x \rightarrow 1} f(x) = \frac{1}{2}$$

$$(1.4) \quad \lim_{x \rightarrow \frac{\pi}{4}} \frac{\tan x}{x} = \frac{\tan \frac{\pi}{4}}{\frac{\pi}{4}} = \frac{1}{\frac{\pi}{4}} = \frac{4}{\pi}$$

Question 2

Given: f is continuous on $\mathbb{R}$, therefore it is also continuous at $x=9$

and therefore: $\lim_{x \rightarrow 9} f(x) = f(9)$

$$f(9) = a$$

$$\lim_{x \rightarrow 9^-} f(x) = \lim_{x \rightarrow 9^-} bx+3 = 9b+3$$

$$\lim_{x \rightarrow 9^+} f(x) = \lim_{x \rightarrow 9^+} \frac{x-3}{\sqrt{x}-3} = \lim_{x \rightarrow 9^+} \sqrt{x}+3 = 6$$

$$\Rightarrow 9b+3 = 6 = f(9) = a$$

$$\Rightarrow a=6 \quad \text{and} \quad 9b=3 \Rightarrow b = \frac{1}{3}$$

Questão 3:

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$$f(x) = \frac{3x^2 + 9x}{x^2 - 9}$$

Horizontal asymptotes:

$$\lim_{x \rightarrow \infty} \frac{3x^2 + 9x}{x^2 - 9}$$

$$= \lim_{x \rightarrow \infty} \frac{3 + 9/x}{1 - 9/x^2} = 3$$

$$\lim_{x \rightarrow -\infty} \frac{3x^2 + 9x}{x^2 - 9} = \lim_{x \rightarrow -\infty} \frac{3 + 9/x}{1 - 9/x^2} = 3$$

$\Rightarrow y = 3$ is a horizontal asymptote.

$$f(x) = \frac{3x(x+3)}{(x-3)(x+3)} = \frac{3x}{x-3}$$

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} \frac{3x}{x-3} = -\infty$$

$$\lim_{x \rightarrow -3} f(x) = \frac{-9}{-6} = \frac{3}{2}$$

$\Rightarrow$ we have 1 vertical asymptote: $x = 3$.

(4) $f(x) = \frac{1}{x+1}$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{1}{x+h+1} - \frac{1}{x+1}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{x+1 - (x+h+1)}{(x+1)(x+h+1)}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{x+1-x-h-1}{h(x+1)(x+h+1)}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{-h}{h(x+1)(x+h+1)}$$

$$= \lim_{h \rightarrow 0} \frac{-1}{(x+1)(x+h+1)}$$

$$= \frac{-1}{(x+1)^2}$$